Journal Pre-proof



Updated Staging System to Define Cardiogenic Shock Severity A Cardiogenic Shock Working Group Report

Van-Khue Ton MD PhD , Song Li MD , Elric Zweck MD , Qiuyue Kong MSc , Shashank S. Sinha MD MSc , Manreet K. Kanwar MD , Jaime Hernandez-Montfort MD MSc , A. Reshad Garan MD , Rachna Kataria MD , Jacob Abraham MD , Paavni Sangal MPH , Borui Li MA , Claudius Mahr MD , William H. Rice MD , Shawn T. Terryah MPH , CSWG Academic Research Consortium, Robert W. Yeh MD , Jacob Møller MD PhD , Daniel Burkhoff MD PhD , Navin K. Kapur MD

PII: S1071-9164(26)00634-2
DOI: https://doi.org/10.1016/j.cardfail.2026.10.004
Reference: YJCAF 6267

To appear in: *Journal of Cardiac Failure*

Received date: 20 August 2026
Revised date: 26 October 2026
Accepted date: 26 October 2026

Please cite this article as: Van-Khue Ton MD PhD , Song Li MD , Elric Zweck MD , Qiuyue Kong MSc , Shashank S. Sinha MD MSc , Manreet K. Kanwar MD , Jaime Hernandez-Montfort MD MSc , A. Reshad Garan MD , Rachna Kataria MD , Jacob Abraham MD , Paavni Sangal MPH , Borui Li MA , Claudius Mahr MD , William H. Rice MD , Shawn T. Terryah MPH , CSWG Academic Research Consortium, Robert W. Yeh MD , Jacob Møller MD PhD , Daniel Burkhoff MD PhD , Navin K. Kapur MD , Updated Staging System to Define Cardiogenic Shock Severity A Cardiogenic Shock Working Group Report, *Journal of Cardiac Failure* (2026), doi: https://doi.org/10.1016/j.cardfail.2026.10.004

This is a PDF of an article that has undergone enhancements after acceptance, such as the addition of a cover page and metadata, and formatting for readability. This version will undergo additional copyediting, typesetting and review before it is published in its final form. As such, this version is no longer the Accepted Manuscript, but it is not yet the definitive Version of Record; we are providing this early version to give early visibility of the article. Please note that Elsevier's sharing policy for the Published Journal Article applies to this version, see: https://www.elsevier.com/about/policies-and-standards/sharing#4-published-journal-article. Please also note that, during the production process, errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

© 2026 Published by Elsevier Inc.

Journal Pre-proof

Updated Staging System to Define Cardiogenic Shock Severity
A Cardiogenic Shock Working Group Report

Van-Khue Ton, MD PhD[1,*], Song Li, MD[2,*], Elric Zweck, MD[3,*], Qiuyue Kong, MSc[4,*], Shashank S. Sinha, MD MSc[5], Manreet K. Kanwar, MD[6], Jaime Hernandez-Montfort, MD MSc[7], A. Reshad Garan, MD[8], Rachna Kataria, MD[4], Jacob Abraham, MD[9], Paavni Sangal, MPH[4], Borui Li, MA[4], Claudius Mahr, MD[2], William H. Rice, MD[2], Shawn T. Terryah, MPH[2], CSWG Academic Research Consortium**, Robert W. Yeh, MD[8], Jacob Møller, MD PhD[10], Daniel Burkhoff, MD PhD[11], Navin K. Kapur, MD[4]

**Equal contribution*

***CSWG ARC authors are listed in the Appendix (Online Supplements)*

Affiliations:

1. The Mass General Brigham Heart and Vascular Institute, Massachusetts General Hospital, Boston, MA, USA
2. Institute for Advanced Cardiac Care, Medical City Healthcare, Dallas, TX, USA
3. Department of Cardiology, Pulmonology, and Vascular Medicine, Medical Faculty and University Hospital Düsseldorf, Heinrich-Heine-University, Düsseldorf, Germany
4. The Cardiovascular Center, Tufts Medical Center, Boston, MA, USA
5. Inova Schar Heart and Vascular, Inova Fairfax Hospital, Falls Church, VA, USA
6. University of Chicago, Chicago, IL, USA
7. Warren Clinic, St. Francis Health, Tulsa, OK, USA
8. Beth Israel Deaconess Medical Center, Boston, MA, USA
9. Center for Cardiovascular Analytics, Research, & Data Science (CARDS), Providence St. Joseph Research Network, Portland, OR, USA
10. Heart Center Copenhagen University Hospital Rigshospitalet and Department of Cardiology, Odense University Hospital, Denmark
11. Cardiovascular Research Foundation, New York, NY, USA

Short title: Data-Driven Updates of the CSWG Shock Stages

Corresponding author: Navin K. Kapur, MD, the CardioVascular Center, Tufts Medical Center, Boston, Massachusetts, USA. Email: Nkapur@tuftsmedicalcenter.org. Phone: 617-636-5000. Fax: 617-636-1444. Twitter handle:@navinkapur4

ACKNOWLEDGEMENTS

The authors thank Colleen Allen for her tireless and outstanding contribution as the Research Project Coordinator for the Cardiogenic Shock Working Group.

Short tweet: Data-driven #CSWG stages. Designed for bedside clinicians. Revised and updated. Largest validation of any staging system in ~1.5 million patients. Check it out!

ABSTRACT

Background: Cardiogenic shock (CS) severity is defined by the Society for Cardiovascular Angiography and Interventions (SCAI) stages and modifications from the Cardiogenic Shock Working Group (CSWG). Lactate thresholds, treatment intensity, and cardiac arrest modifiers remain incompletely defined. We aimed to refine the 2022 CSWG staging criteria using contemporary data.

Methods: Patients with CS from heart failure (HF-CS) or myocardial infarction (MI-CS) enrolled in the multicenter CSWG registry (2021-2025) were analyzed. Associations between systolic blood pressure (SBP), lactate, treatment intensity, and in-hospital mortality, which is considered a surrogate for CS severity, were examined. External application of the updated system was performed in an independent U.S. health system.

Results: Among 11,640 patients (60.5% HF-CS, 26.1% MI-CS), SBP and lactate demonstrated graded relationships with mortality across thresholds (SBP<60, 60-<90, ≥90 mmHg; lactate 0-2, >2-5, >5-10, >10 mmol/L). Mortality increased incrementally with treatment intensity. The 2026 staging system incorporating these thresholds and the number of vasoactive drugs and temporary mechanical circulatory support devices (stage C: 1-2, D: 3-4, E: ≥5) demonstrated progressive mortality across stages. Compared with prior CSWG staging, 28.4% of stage D and 56.4% of stage E patients were reclassified, with lower mortality among those reclassified to less severe stages. Out-of-hospital cardiac arrest functioned best as a stage modifier, whereas in-hospital cardiac arrest corresponded to stage E. Findings were consistent across HF-CS and MI-CS. The 2026 CS staging system was evaluated in 1,417,637 patients in a geographically and structurally diverse healthcare system and demonstrated a graded, monotonic increase in mortality across

Journal Pre-proof

Journal Pre-proof

stages A-E.

Conclusions: Using fewer clinical variables, the 2026 CSWG criteria provide a simple framework for CS severity staging that is applicable among HF-CS and MI-CS patients, and can be standardized across clinical trials and healthcare settings.

KEY WORDS: Cardiogenic shock stages, severity, serial assessment, shock progression, mortality

ABBREVIATIONS:

CS: cardiogenic shock

tMCS: temporary mechanical circulatory support

INTRODUCTION

Mortality for patients with cardiogenic shock (CS) remains high[1]. Heterogeneity in presentation and etiologies likely contributes to different treatment effects and adverse patient outcomes. In the United States, CS from heart failure (HF-CS) is increasingly more common than myocardial infarction-related CS (MI-CS)[2,3]. In 2019, the Society for Cardiovascular Angiography and Interventions (SCAI) proposed a clinical staging system of CS severity based on consensus opinion[4]. Cardiac arrest was later added as a stage E risk modifier of mortality[5]. In 2022, the Cardiogenic Shock Working Group (CSWG) added specific thresholds for systolic blood pressure (SBP), mean arterial pressure (MAP), lactate, pH, alanine aminotransferase (ALT), and treatment intensity for each stage, in order to more consistently characterize patients across specialties and diverse clinician perspectives[6]. Using in-hospital mortality as a surrogate

for CS severity, the 2019 SCAI and 2022 CSWG systems have been retrospectively validated, showing incremental increase in mortality with advancing stages, with mortality exceeding 50% among stage E patients[5,7-11]. We previously demonstrated that CSWG classification can serially and reproducibly assess CS severity[11,12]. CSWG stages have also been utilized in the DanGer Shock and AltShock2 randomized controlled trials[13,14].

Despite widespread adoption, both the SCAI classification and its modification by CSWG have limitations. The former, largely based on expert consensus, can be interpreted subjectively and did not distinguish between out-of-hospital (OHCA) vs. in-hospital cardiac arrest (IHCA). The 2022 CSWG modification, conducted in 3,455 patients, had lactate thresholds that did not capture all clinical scenarios. Automatic assignment to stage E for OHCA did not permit serial severity assessment. It is unknown if SBP, MAP, pH, and ALT remained valid markers of disease severity in a much larger, more contemporary CS cohort. There is an unmet need for refinement of the pre-existing CS severity staging system with unambiguous cutoffs, to provide a unifying framework for CS studies across all healthcare settings and clinical trials, while maintaining the original intent of the SCAI stages: a simple schema easily deployed at the bedside. There is increasing recognition that HF-CS and MI-CS have distinct clinical presentations, trajectories, and outcomes, suggesting that etiology-specific risk staging may be warranted.

In this study, we aimed to: (1) systematically analyze patients' data to test **validity of components** of the 2022 CSWG stages over the first 72 hours post CS diagnosis, using a much larger CSWG cohort; (2) provide **observed** mortality rates for each stage among patients with HF-CS or MI-CS; (3) compare mortality stratification and CS stage reclassification between the 2022 vs. the 2026 system; (4) assess the impact of both OHCA and IHCA on risk-staging; (5)

Journal Pre-proof

externally apply the CSWG stages in an independent, large U.S. health system database; and (6) better define outcomes associated with stage A (at risk for CS). Notably, the **primary aim** of this study was to **refine** the 2022 CSWG system (i.e. not to derive a comprehensive mortality-predicting tool), and show that the updated CS severity stages, with fewer clinical variables and more data-driven thresholds, could still demonstrate a gradient of observed mortality rates from stage A to E CS.

METHODS

Data Source

The CSWG is an academic consortium of 36 international (majority from the U.S.) CS centers, with Institutional Review Board approval provided at each site. Details of the registry have been published previously[6,15]. Briefly, the registry contains retrospectively collected data defined *a priori* by the principal investigators. CS diagnosis was physician-adjudicated at each site and defined as: SBP <90 mmHg for at least 30 minutes, cardiac index <2.2 L/min/m^2, or the need for vasoactive drugs and tMCS devices. Treatment strategies specifically for CS were determined by treating clinicians. Data quality was ensured through centralized auditing and adjudication via annual site visits and random chart queries by CSWG staff. CS patients enrolled between 2021 and 2025 were included. CS etiology was due to MI or HF. Other CS etiologies were excluded in this analysis. SBP, MAP, lactate and treatment intensity were collected on admission to CSWG sites (baseline) and subsequently every 6 hours through 72 hours after CS diagnosis. Vasoactive-inotrope score (VIS) was calculated as previously shown[16]. The primary outcome was unadjusted in-hospital mortality.

The updated CSWG stages were applied to patients admitted to hospitals of the largest U.S. health system (HCA Healthcare, Nashville, TN). Patients admitted at 161 HCA hospitals in

20 U.S. states (2021-2024) were identified from electronic medical records using a combination of: *i.* age ≥18 years, *ii.* having ≥1 HF or MI-related ICD-10 codes, or R570.x (CS) in the final discharge diagnosis list, and *iii.* meeting CSWG stage A-E criteria at any point during hospitalization. SBP and lactate were obtained from electronic chart queries. A list of ICD-10 codes used to identify patients is in **Table S1**. Since most patients admitted for MI or HF did not have CS, we did not use R570.x as the primary diagnostic code, but instead included all possible codes for MI or HF-related diagnoses, to capture patients with CS who would be classified as stage A. Therefore, patients were only included if one or more of these codes were in the top 10 final discharge diagnoses. There was no overlap between CSWG registry and HCA dataset.

Statistical Analysis

Continuous variables were reported as mean±standard deviations or median (interquartile range). Comparisons of continuous variables were performed using the student's t-test or the Wilcoxon test as appropriate. Categorical variables were compared using chi-square tests. Locally weighted scatterplot smooth (LOWESS) plots and linear regression were used to assess correlations between tested variables and mortality. All mortality rates were observed rates (i.e. [number of dead patients/number of alive patients]x100). The likelihood-ratio test was used to compare model fit. Data missingness for the CSWG and HCA cohorts is shown in **Table S2**. SBP and lactate thresholds were chosen based on thresholds used in our previous analysis, and assessed for correlation with mortality using linear regression[6]. The updated criteria were applied to the HCA Healthcare dataset on admission for CS, and throughout hospitalization for maximum CS stages.

To address missing baseline data, we performed a sensitivity study where continuous (SBP, lactate, pH, ALT) and categorical (tMCS device count, vasoactive drug count, OHCA)

variables were first imputed using multiple imputation by chained equations (MICE) with m=5 imputations. CS stages were subsequently derived within each completed dataset, with final statistical analyses pooled using Rubin's rules.

Statistical significance was defined as a two-sided alpha of 0.05. Analyses were performed using SAS v.9.4 (SAS Institute Inc., Cary, NC).

RESULTS

Baseline characteristics

Of 15,095 patients, 11,640 patients (77.1%) in the most recent versions of CSWG registry (2021-2025) were included (22.9% belonged to the earlier CSWG versions and were not used). Of the included patients, 7,039 (60.5%) had HF-CS, and 3,040 (26.1%) had MI-CS (**Table 1**). Compared to patients with MI-CS, those with HF-CS were younger (62 [51-70] vs. 66 [58-74] years, p<0.001), more likely to be Black (22.6% vs. 7.7%, p<0.001), have atrial fibrillation or atrial flutter (40.7% vs. 12.8%, p<0.001), chronic kidney disease (36.3% vs. 17.8%, p<0.001), and implantable cardioverter-defibrillator (31.5% vs. 4.8%, p<0.001). Other baseline characteristics, laboratory values, vital signs and hemodynamics are shown in **Table 1**.

Determination of threshold values

The clinical variables and thresholds used in 2022 CSWG staging system (SBP, MAP, lactate, ALT, pH, vasoactive drugs, tMCS devices) were selected by CSWG "expert consensus opinion […] based on recently published reports evaluating these variables and their association with SCAI stages"[6]. Using a larger, more contemporary CSWG cohort, we evaluated the association between these same variables with in-hospital mortality, which was used as a surrogate for CS severity. We refined thresholds of the selected variables for each CS stage, and assessed whether fewer variables could yield a similarly robust CS staging system.

Baseline SBP, MAP, ALT, pH and lactate

LOWESS plots showed relationships between baseline variables and in-hospital mortality in patients with HF-CS and MI-CS (**Figure 1**). SBP, MAP, and pH were inversely correlated with mortality, while lactate showed a positive linear correlation. ALT showed minimal association with mortality, even when limited to ≤1000 U/L, suggesting that it could be removed from the staging criteria. Patterns were similar between HF-CS and MI-CS, supporting a single classification scheme for both. We elected to retain SBP instead of MAP, and lactate instead of pH in keeping with CS-defining criteria used in prior CS randomized controlled trials[13,17].

Given the near linear association between SBP or lactate to mortality, we opted to keep the same thresholds validated in the 2022 CS stages, acknowledging that the threshold choices may be arbitrary. The thresholds were: SBP <60, 60 - <90, ≥90 mmHg, lactate 0 - 2, >2 - 5, >5 - 10, >10 mmol/L. SBP and lactate values were serially obtained on admission to CSWG sites (baseline), and at 6, 12, 18, 24, 48, and 72 hours post CS diagnosis. Mortality linearly increased with each worsening of SBP and lactate thresholds at all time points and exceeded 50% at SBP <60 mmHg and lactate >10mmol/L (**Figure 2A-B**).

For patients on no vasoactive drugs and tMCS devices, mortality also rose linearly with lactate and SBP as continuous variables or thresholds at all time points (**Figure S1**), supporting the continued use of these lactate and SBP cutoffs in the updated system.

Treatment intensity

The number of vasoactive drugs and tMCS devices was serially collected on admission, and every 6 hours post CS diagnosis to 72 hours. Mortality rose in proportion to the number of drugs and devices when counted individually (**Figure 2C-D**), or combined (**Figure 2E**). We selected the total number of simultaneously used drugs and devices of 5 to be the threshold for

stage E when mortality exceeded 50% as previously shown in studies using the SCAI classification[7,9] (**Figure 2E**).

We considered stage C patients as having "hypotension and mildly elevated lactate on no treatments", or "classic CS responsive to treatments", while stage D patients were considered to have "deteriorating CS despite treatments". Therefore, patients on 1-2 drugs and devices with SBP ≥90 mmHg and lactate 0 - 2 mmol/L were grouped in stage C. This approach also accounted for patients transferred to CSWG sites already on drugs and tMCS. Patients on 1-2 drugs and devices with SBP of 60 - <90 mmHg or lactate of >2 - 10 mmol/L were classified as stage D due to having persistent hypotension or lactate elevation despite treatments.

Vasoactive-inotrope scores (VIS)

Among patients with available data, VIS progressively increased with worsening CS stages (C: 2.5 [0, 5], D: 5 [0, 20], E: 20 [0, 406.8]) (**Table S3**). Adding VIS to SBP, lactate and treatment intensity did not significantly improve the overall model fit by the likelihood-ratio test (p=0.159), suggesting that VIS provided limited incremental value for predicting mortality after accounting for the number of drugs and devices.

Updates to the CSWG staging criteria

Based on these data, the following updates to the 2022 CSWG classification were incorporated:

1. SBP and lactate were used, omitting ALT, MAP, and pH.

2. Lactate threshold of >5 - 10 mmol/L was added to stage D for patients on no drugs and tMCS devices, encompassing clinical scenarios missed by the 2022 CSWG system.

3. The total number of drugs and devices incrementally increased with stage worsening (C: 1-2, D: 3-4, E: ≥5).

Journal Pre-proof

4. Removal of OHCA as an automatic criterion for stage E.

The SAS code for CSWG stage assignment is provided in the Appendix (Supplemental Materials). A summary of changes from the previous CSWG criteria is shown in **Table 2**.

Stratification of in-hospital mortality by CS stages

Unadjusted in-hospital mortality was lower among the HF-CS cohort (24.2% vs. 37.1%, p<0.001). The updated criteria for CS stages were applied to the CSWG cohort. Most patients changed their stages by 6 hours post CS diagnosis (**Figure 3A**). In-hospital mortality was highest among patients with stage E, followed by stage D at all time points (**Figure 3B, left panel**) (overall p<0.001). Mortality trends over all time-points were largely similar among the HF-CS and MI-CS cohorts (**Figure S2**).

Comparison between the 2022 and 2026 CSWG stages

To demonstrate the incremental benefit of the updated 2026 stages, we compared them to the 2022 CS stages. In the 2026 stages, few patients were reclassified from stage B to A (9.9%) or C to B (2.0%) (**Table 3**). In contrast, 28.4% stage D and 56.4% stage E patients were reclassified. Mortality of those reclassified to stages A-C CS was lower than those remaining in stage D or E (**Table 3**). Notably, 2.4% of patients unable to be classified in the 2022 system could now be assigned a stage in the updated system. Using fewer variables, the 2026 system showed mortality gradient across the stages, similarly to the 2022 system (**Figure 3B, right panel**). Mortality rates across baseline stages A to E were very similar after multiple imputation was performed to account for data missingness (**Table S4**).

External application of the updated CSWG stages

All patients

The 2026 CSWG stages were applied in 1,417,637 patients from HCA Healthcare.

Mortality incrementally increased from baseline stage A (4.8%), B (14.0%), C (21.9%), D (34.1%), to E (59.8%) (**Figure 4A, left panel)**. Similar trends were seen in HF-CS (baseline A: 3.4%, B: 9.3%, C: 15.3%, D: 24.6%, E: 44.8%) and MI-CS (baseline A: 6.5%, B: 19.0%, C: 25.5%, D: 36.1%, E: 58.4%) cohorts (**Figure S4A**).

Mortality stratification across baseline stages were similar between the 2026 and 2022 systems (**Figure 4A, left vs. right panel**), with fewer variables in the 2026 system.

For patients admitted to HCA Healthcare hospitals in different geographic locations (**Figure 4B**) or hospitals of small to large volumes (**Figure 4C**), mortality also increased incrementally across baseline CS stages, suggesting that the 2026 classification may be applicable in diverse clinical settings.

CS trajectory

Regarding CS stage worsening, stage C patients had the highest proportion that progressed (60.1%), followed by stage B (30.1%) and D (25.3%), while only a minority (7.7%) of baseline stage A progressed to maximum stages C-E throughout hospitalization (**Figure 4D**). Mortality incrementally increased across the maximum stages in patients who progressed to more severe CS (**Figure 4E**). Among subjects presenting with no CS (stage A) or stage B-D CS, those who progressed to stage E had similarly high mortality to those who had stage E CS on admission. CS stage progression among HF-CS and MI-CS groups was similar (**Figure S4B-C**).

Mortality of stage B vs. stage C patients

In the CSWG cohort, stage B patients at baseline had slightly higher observed mortality than stage C patients under both the 2026 (25.0% vs. 20.5%) and 2022 (24.8% vs. 17.4%) staging systems. In contrast, baseline stage B patients in the HCA dataset had lower mortality than stage C patients (14.0% vs. 21.9%). Specifically, stage B patients in HCA with isolated

hypoperfusion (lactate >2-5 mmol/L) or isolated hypotension (SBP 60 - <90 mmHg) had lower mortality than stage C patients with combined hypoperfusion and hypotension (13.6% vs 16.2% vs. 21.9%, respectively) (**Figure 4F**). Mortality in both stage B and C groups remained substantially higher than in stage A patients (4.8%). MI-CS and HF-CS patients exhibited similar mortality trends among stage B subgroups vs. stage C.

The difference in stage B vs. C mortality rates between CSWG and HCA datasets likely reflects selection bias in CSWG enrollment: stage B patients who remained stable were less likely to be enrolled in the CSWG registry which relied on manual enrollment, whereas those who deteriorated were more likely to be identified and enrolled. Therefore, less severely ill stage B patients were likely underrepresented, potentially inflating the observed mortality rate in this group. In the HCA dataset, patients were identified automatically through an electronic medical record query, eliminating enrollment bias. As a result, stage B mortality in the HCA dataset was less than that in the CSWG dataset and in between stages A and C in the HCA dataset.

CS stages among non-transferred CSWG patients

We applied the updated criteria to a subset of patients admitted directly to CSWG sites (non-transferred patients) (**Figure S4**). Proportions and mortality of patients in each stage over time mirrored those of patients in the entire CSWG population.

Staging of patients with cardiac arrest in CSWG cohort

Mortality over all time points was higher among CSWG patients in baseline stages B-E with OHCA than those without cardiac arrest (**Figure 5A**). Specifically, the relative increase in mortality was 58.7% for patients with baseline stage B, 18.4% for stage C, 37.8% for stage D, and 18.7% for stage E, suggesting that OHCA should be a stage modifier. In contrast, mortality was markedly elevated, often exceeding 50% in patients with IHCA across all CSWG stages,

indicating that IHCA should have an automatic stage E designation (**Figure 5B**).

DISCUSSION

In this study, we refined CS severity staging criteria of the 2022 CSWG system, using data from the largest international CS registry to date and applied in the largest series of patients from HCA Healthcare database. Notably, this updated 2026 shock severity classification system was intentionally designed to be simple, pragmatic, and applicable across all levels of clinical expertise and hospital settings. We did not set out to derive a model with maximal predictive power for in-hospital mortality. Rather, we prioritized usability by incorporating three readily available clinical variables: SBP, lactate, and treatment intensity agnostic of specific drugs and tMCS devices. Although a more complex model, such as one based on machine learning, could likely achieve a higher discriminative performance, such approach is difficult to implement consistently and quickly in routine clinical practice across diverse healthcare settings. The updated shock severity stages are intended to provide a practical framework and a unified language to facilitate communication, risk stratification, and patient classification. They are not intended to replace clinical judgment or serve as the sole determinant of shock severity, but rather to complement comprehensive clinical assessment. Future prospective studies are required to further establish the clinical utility of CS staging systems. The main findings were (**Central Illustration**):

1. The updated 2026 CS staging system stratified mortality across the stages, similar to the 2022 CSWG system, but using fewer clinical parameters. Variable thresholds now encompassed all clinical scenarios, thereby meaningfully enhancing bedside applicability as compared to prior frameworks.

2. This single staging system can be applied to both HF-CS and MI-CS cohorts.

3. OHCA functioned best as a risk modifier rather than a stage-defining variable, while IHCA should be an automatic stage E.

The choice of variables and their thresholds

Traditionally, SBP <90 mmHg has been used to define CS since the SHOCK trial[17] and continued in contemporary CS trials[14,18,19], and lactate ≥2.5 mmol/L was used to define CS in DanGer Shock[13], so we retained SBP and lactate for consistency with prior literature. Although LOWESS plots showed linear relationships between SBP or lactate and mortality, making their thresholds somewhat arbitrary, the selected cut-off values were validated in the 2022 CSWG system and again in this study, demonstrating mortality gradients across the stages.

We chose the total number of simultaneously used vasoactive drugs and tMCS devices as measures of treatment intensity instead of other parameters such as VIS, hemodynamics, or specific tMCS devices, for the following reasons: *i.* VIS is often not easily nor rapidly calculated at the bedside, and did not meaningfully enhance model fit, *ii*. Hemodynamics obtained from a pulmonary artery catheter may not be performed at all hospitals, *iii*. The specific choices of drugs or tMCS devices, as well as medication doses and timing of device escalation were not standardized and dependent on the treating teams' preferences and resource availability, *iv*. The cumulative burden of drugs and devices is strongly correlated with in-hospital mortality and provided mortality gradients among the stages. Thus, the use of simple variables (SBP, lactate), together with the total number of vasoactive drugs and tMCS devices to assign CS stages, agnostic of specific drugs or devices, produced a system that correlated with mortality, and can be rapidly and serially applied at the bedside across diverse healthcare settings.

The inclusion of patients on 1-2 drugs/devices with SBP ≥90 mmHg and lactate 0 - 2 mmol/L to stage C should only apply to those with the specific diagnosis of CS (SBP <90 mmHg

for at least 30 minutes, cardiac index <2.2 L/min/m^2, or need for vasoactive drugs/tMCS devices[6]). Patients with HF treated empirically with inotropes without evidence of CS, or those receiving a microaxial flow pump for high-risk coronary interventions, should not be included.

Stage B remains bad

In the 2022 analysis, patients classified as CSWG stage B exhibited slightly higher in-hospital mortality than those in stage C[6]. This observation was attributed to the inclusion of "hypoperfusion" within stage B (hypotension *or* hypoperfusion), which differs from the original 2019 and revised 2022 SCAI definition of stage B as "hypotension without hypoperfusion"[4]. The CSWG's expanded stage B criterion was intended to capture a vulnerable subgroup of HF-CS patients with normotensive shock but evidence of mildly elevated lactate as surrogate for the beginning of hypoperfusion.

A closer look at sub-phenotypes among stage B and C patients was revealing. Stage B patients with isolated hypotension or isolated, mild lactate elevation had lower mortality than stage C patients with hypotension *and* elevated lactate. This was consistent with prior studies in non-CSWG registries[9,20]. Notably, enrollment in the CSWG registry is likely biased against patients presenting with baseline stage A (at risk) or stage B CS who never progress to more advanced stages. Such patients, who by definition do not receive vasoactive medications or tMCS devices, are substantially less likely to be identified and enrolled. In contrast, patients treated with drugs and/or devices are typically managed in intensive care or step-down units, where they are more readily identified and recruited into the registry. More than half of the patients in the CSWG registry were transferred from outside institutions[21], most of whom were already receiving vasoactive therapies or tMCS at the time of arrival. Consequently, patients receiving no drugs or devices were likely underrepresented in our registry, resulting in an

artificially elevated observed mortality rate in our stage B cohort. In contrast, in the external validation cohort of nearly 1.5 million patients from HCA Healthcare, stage B patients were identified objectively and comprehensively using electronic medical record data and demonstrated **significantly lower mortality** than stage C patients, whether they were treated at larger or smaller hospitals in diverse geographical locations. These findings indicate that the updated CSWG criteria reproduce the expected stepwise increase in mortality across CS stages in a large, real-world population. Additionally, while it is not surprising that 60% of patients with stage C (i.e. "classic" CS) at baseline progressed to stages D-E during hospitalization, almost 1 in 3 stage B patients progressed to stages C-E CS as well, highlighting the increased risk of adverse outcomes even in "beginning" CS.

Time is tissue

In the validation cohort of HCA Healthcare, patients who presented with no or less severe CS but deteriorated had higher mortality than those who stabilized. Mortality of those progressing to stage E later during hospitalization was similar to or higher than that of subjects presenting in stage E CS on admission. This finding is consistent with prior results from a smaller CSWG cohort[6], but now identifies that most stage A patients - "at risk" on admission without shock - are unlikely to deteriorate. However, if stage A subjects do progress to stages B-E CS, their mortality rates are comparable to those presenting with stages B-E CS at baseline.

The precise definition of hypoperfusion

It is worth noting that "hypoperfusion" in CSWG stages is specifically defined as "lactate >2 mmol/L", and lactate elevation alone may not be sufficient to quantify hypoperfusion. Future studies are needed to assess the prognostic values of other perfusion markers, individually or cumulatively, such as urine output[5], central venous-arterial carbon dioxide difference

(PCO_2 gap), or PCO_2 gap/central arterial-to-venous oxygen difference ($C_{a-v}O_2$)[22] in predicting mortality. Physical exam signs of hypoperfusion (mottled skin, cold extremities, altered mental status), as recommended by the 2019 SCAI document[5], should be relied upon only when present, as their absence does not rule out CS.

In and out-of-hospital cardiac arrest modify mortality

In the 2022 CSWG study conducted in 3,455 patients, mortality of patients with OHCA was as high as those in stage E, hence the classification of OHCA into stage E[6]. The present study, conducted in a much larger cohort (11,640 patients), showed higher mortality across all CS stages for subjects with OHCA compared to those without. The increased mortality in the OHCA cohorts might have been due to death from anoxic brain injury[23], although the modes of death are not captured in our registry. Future iterations of CSWG registry that capture causes of death will help answer this hypothesis. Based on our data, we contend that OHCA is now designated a stage modifier, not an automatic stage E assignment.

In contrast, patients with IHCA had mortality exceeding 50% across all stages, indicating that such patients should automatically be designated as stage E. The starkly higher mortality rate among IHCA patients compared to those with OHCA were likely due to survival bias; that is, only survivors of OHCA that survive to hospital admission can be enrolled in the registry. We did not include IHCA as a specific criterion for the updated stages for two reasons: first, IHCA does not have an ICD-10 code, making it difficult to capture in an automated data query; and second, patients with IHCA would most likely meet criteria for stage E due to extremely low SBP and high lactate.

Limitations

Journal Pre-proof

This study is limited by its retrospective design and data missingness that may not be random (**Table S2**); patients with partial data (available SBP but no lactate) were not included in stage calculations, which might have biased the cohort toward sicker, higher-acuity patients with frequent lactate assessments. However, compared to the higher rates of missingness in CSWG registry, the proportions of missingness in HCA Healthcare dataset were quite small. Use of vasoactive agents and tMCS devices was heterogeneous and varied across centers. We did not capture the intent of vasoactive drugs and tMCS deployment, specific causes of cardiac arrest (ventricular arrhythmias, pulseless electrical activity) and causes of death for CS. Lack of standardization in device selection, timing, and escalation of tMCS may limit reproducibility and/or generalizability. Capturing of SBP could have been inaccurate among patients with an intra-aortic balloon pump. Refinement of the updated criteria were performed retrospectively. Future prospective validation studies are needed. Survival bias might have impacted observed mortality rates in CSWG registry and HCA dataset. Residual unmeasured confounding cannot be excluded.

CONCLUSION

Using the largest multicenter CS registry to date and data from the largest U.S. health system, we now report a refinement to the CSWG staging criteria for CS severity that effectively stratified mortality gradients across stages A through E. The updated CSWG stages use simple, easily obtained clinical variables (SBP, lactate, total number of vasoactive drugs and tMCS devices). These key updates now serve as a robust mechanism for identifying CS patients, defining their risk of dying, and studying their trajectory during hospitalization across different healthcare settings, as well as in randomized controlled trials testing new treatment strategies for this devastating condition affecting millions of patients worldwide.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the first author (VKT) used ChatGPT in order to edit texts to be more concise. After using this tool/service, Dr. Ton reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Journal Pre-proof

REFERENCES

1. Thiele H, Hassager C. Cardiogenic Shock. *The New England journal of medicine* 2026; **394**(1): 62-77.
2. Hernandez-Montfort J, Kanwar M, Sinha SS, et al. Clinical Presentation and In-Hospital Trajectory of Heart Failure and Cardiogenic Shock. *JACC Heart failure* 2023; **11**(2): 176-87.
3. Kanwar MK, Billia F, Randhawa V, et al. Heart failure related cardiogenic shock: An ISHLT consensus conference content summary. *The Journal of heart and lung transplantation : the official publication of the International Society for Heart Transplantation* 2023.
4. Baran DA, Grines CL, Bailey S, et al. SCAI clinical expert consensus statement on the classification of cardiogenic shock: This document was endorsed by the American College of Cardiology (ACC), the American Heart Association (AHA), the Society of Critical Care Medicine (SCCM), and the Society of Thoracic Surgeons (STS) in April 2019. *Catheterization and cardiovascular interventions : official journal of the Society for Cardiac Angiography & Interventions* 2019.
5. Naidu SS, Baran DA, Jentzer JC, et al. SCAI SHOCK Stage Classification Expert Consensus Update: A Review and Incorporation of Validation Studies: This statement was endorsed by the American College of Cardiology (ACC), American College of Emergency Physicians (ACEP), American Heart Association (AHA), European Society of Cardiology (ESC) Association for Acute Cardiovascular Care (ACVC), International Society for Heart and Lung Transplantation (ISHLT), Society of Critical Care Medicine (SCCM), and Society of Thoracic Surgeons (STS) in December 2021. *Journal of the American College of Cardiology* 2022; **79**(9): 933-46.
6. Kapur NK, Kanwar M, Sinha SS, et al. Criteria for Defining Stages of Cardiogenic Shock Severity. *Journal of the American College of Cardiology* 2022; **80**(3): 185-98.
7. Jentzer JC, van Diepen S, Barsness GW, et al. Cardiogenic Shock Classification to Predict Mortality in the Cardiac Intensive Care Unit. *Journal of the American College of Cardiology* 2019; **74**(17): 2117-28.
8. John KJ, Stone SM, Zhang Y, et al. Application of Cardiogenic Shock Working Group-defined Society for Cardiovascular Angiography and Interventions (CSWG-SCAI) Staging of Cardiogenic Shock to

the Medical Information Mart for Intensive Care IV (MIMIC-IV) database. *Cardiovascular revascularization medicine : including molecular interventions* 2023; **57**: 82-90.
9. Patel SM, Berg DD, Bohula EA, et al. Continuum of Preshock to Classic Cardiogenic Shock in the Critical Care Cardiology Trials Network Registry. *JACC Heart failure* 2024; **12**(9): 1625-35.
10. Ris TH, Atazadah M, Hoek R, et al. Predictive value of the cardiogenic shock working group-modified SCAI criteria in early-stage heart failure-related cardiogenic shock. *International journal of cardiology Heart & vasculature* 2025; **60**: 101776.
11. Zweck E, Ton VK, Kanwar M, et al. CSWG-SCAI Stages Combined With Machine Learning-Based Phenotypes for Serial Risk Stratification in Cardiogenic Shock. *JACC Heart failure* 2025; **13**(10): 102611.
12. Ton VK, Li S, John K, et al. Serial Shock Severity Assessment Within 72 Hours After Diagnosis: A Cardiogenic Shock Working Group Report. *Journal of the American College of Cardiology* 2024.
13. Møller JE, Engstrøm T, Jensen LO, et al. Microaxial Flow Pump or Standard Care in Infarct-Related Cardiogenic Shock. *The New England journal of medicine* 2024; **390**(15): 1382-93.
14. Morici N, Sacco A, Frea S, et al. Early Intra-Aortic Balloon Support for Heart Failure-Related Cardiogenic Shock: A Randomized Clinical Trial. *Journal of the American College of Cardiology* 2025; **85**(16): 1587-97.
15. Thayer KL, Zweck E, Ayouty M, et al. Invasive Hemodynamic Assessment and Classification of In-Hospital Mortality Risk Among Patients With Cardiogenic Shock. *Circulation Heart failure* 2020; **13**(9): e007099.
16. Vallabhajosyula S, Faugno AJ, Li B, et al. Prognostic Implications of Quantifying Vasoactive Medications in Cardiogenic Shock. *Journal of cardiac failure* 2024; **30**(11): 1516-21.
17. Hochman JS, Sleeper LA, Webb JG, et al. Early revascularization in acute myocardial infarction complicated by cardiogenic shock. SHOCK Investigators. Should We Emergently Revascularize Occluded Coronaries for Cardiogenic Shock. *The New England journal of medicine* 1999; **341**(9): 625-34.
18. Mathew R, Di Santo P, Jung RG, et al. Milrinone as Compared with Dobutamine in the Treatment of Cardiogenic Shock. *The New England journal of medicine* 2021; **385**(6): 516-25.
19. Thiele H, Zeymer U, Akin I, et al. Extracorporeal Life Support in Infarct-Related Cardiogenic Shock. *The New England journal of medicine* 2023; **389**(14): 1286-97.
20. Jentzer JC, Burstein B, Van Diepen S, et al. Defining Shock and Preshock for Mortality Risk Stratification in Cardiac Intensive Care Unit Patients. *Circulation Heart failure* 2021; **14**(1): e007678.
21. Garan AR, Kataria R, Li B, et al. Outcomes of Patients Transferred to Tertiary Care Centers for Treatment of Cardiogenic Shock: A Cardiogenic Shock Working Group Analysis. *Journal of cardiac failure* 2024; **30**(4): 564-75.
22. Janotka M, Ostadal P. Biochemical markers for clinical monitoring of tissue perfusion. *Molecular and cellular biochemistry* 2021; **476**(3): 1313-26.
23. Berg DD, Singal S, Palazzolo M, et al. Modes of Death in Patients with Cardiogenic Shock in the Cardiac Intensive Care Unit: A Report from the Critical Care Cardiology Trials Network. *Journal of cardiac failure* 2024; **30**(5): 728-33.

Journal Pre-proof

FIGURE LEGENDS

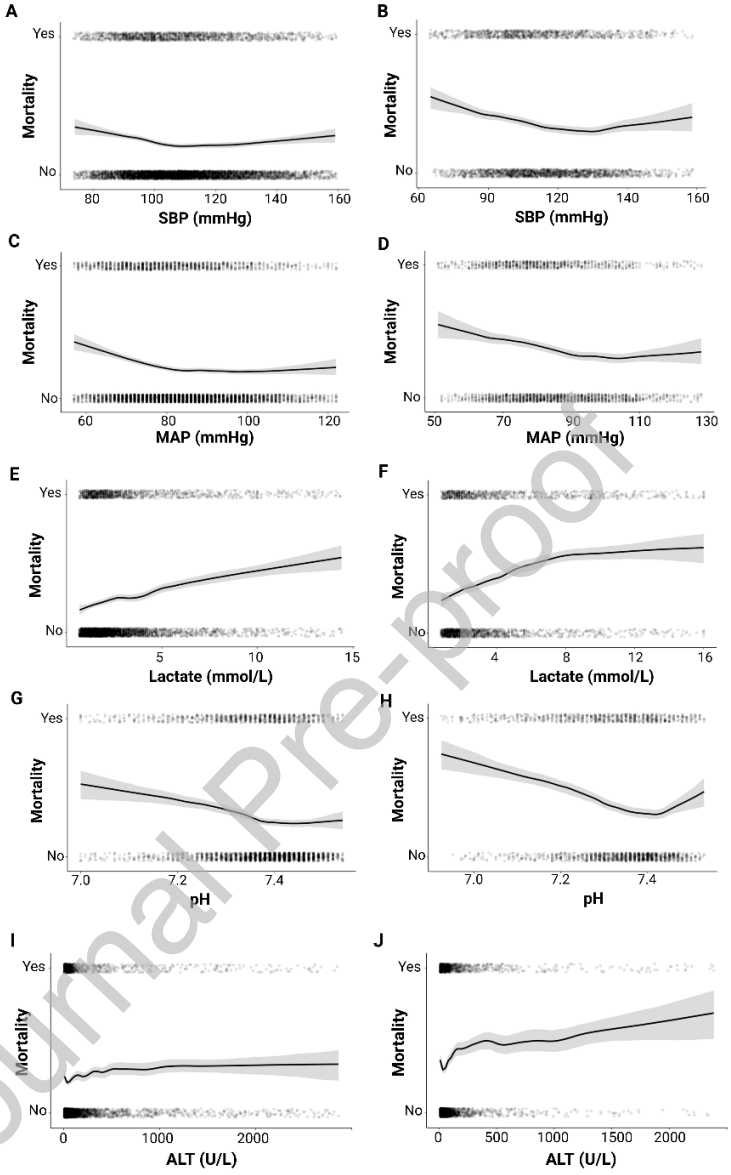
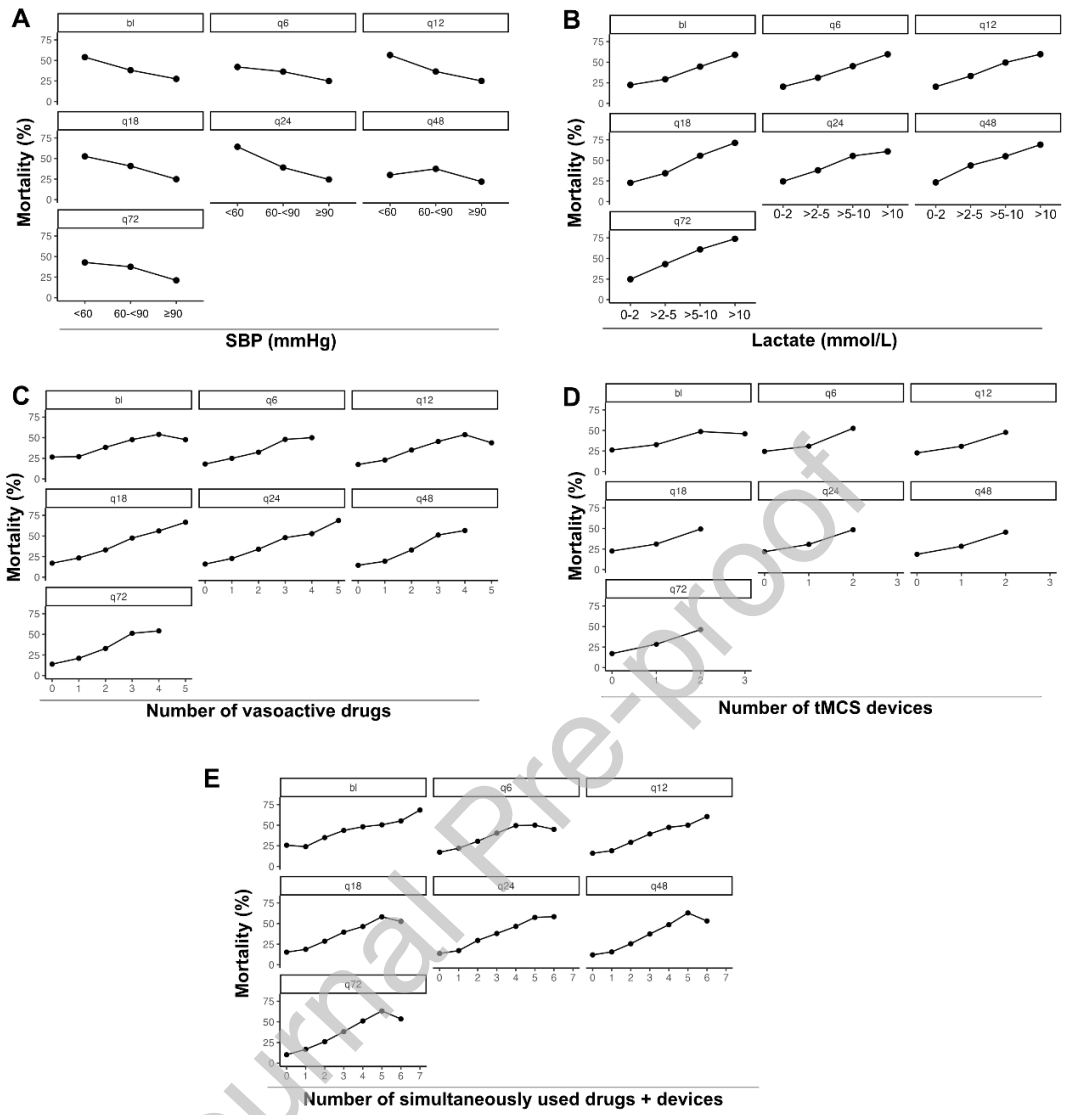


Figure 1: LOWESS plots showing correlations with in-hospital mortality. Left column: HF-CS; right column: MI-CS. **A-B.** Systolic blood pressure (SBP). **C-D.** Mean arterial pressure (MAP). **E-F.** Lactate. **G-H.** pH. **I-J.** Alanine transaminase (ALT).

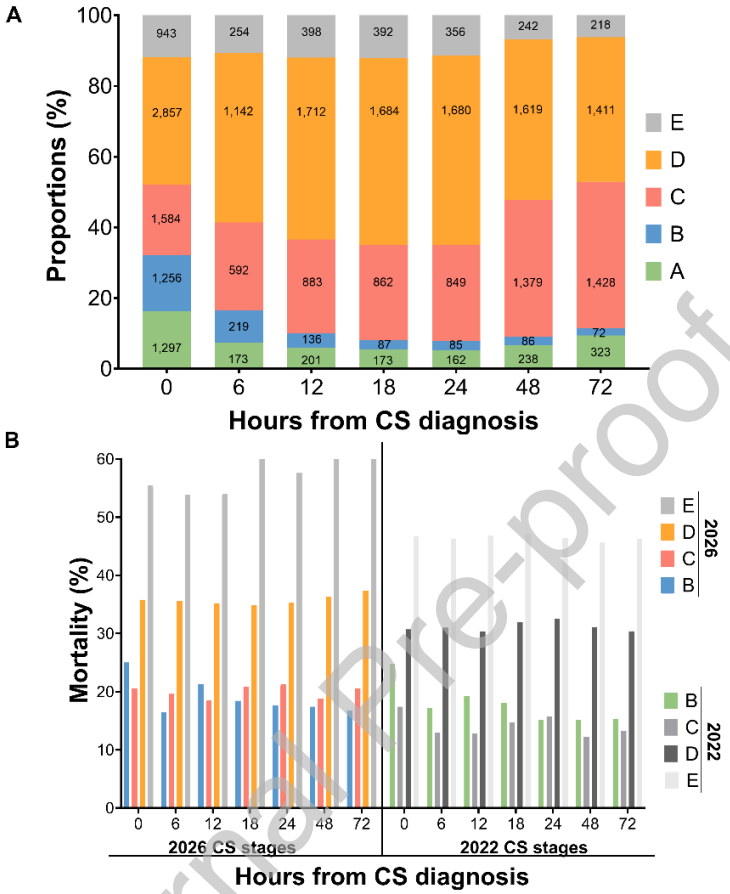
Journal Pre-proof



Journal Pre-proof

Figure 2: Determination of threshold values. Linear regression between SBP and lactate thresholds or treatment intensity and mortality over time. **A.** Systolic blood pressure (SBP). **B.** Lactate. **C.** Number of vasoactive drugs. **D.** Number of tMCS devices. **E.** Treatment intensity

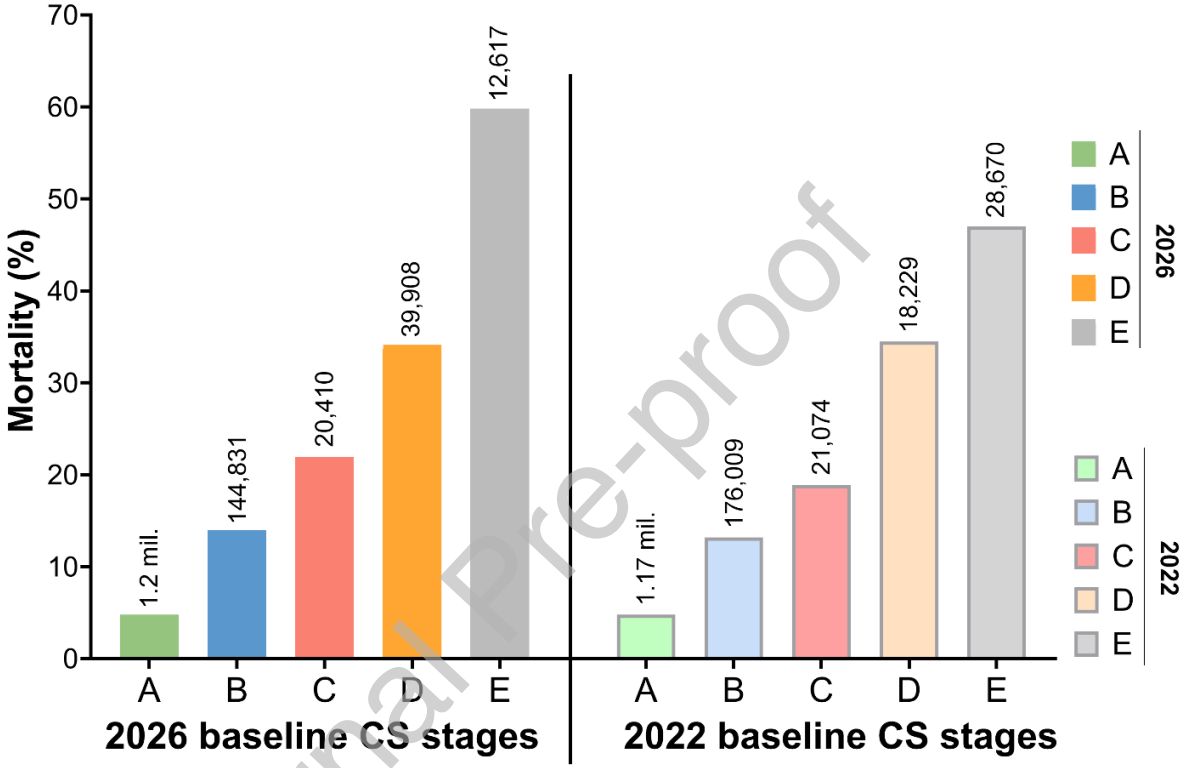
(total number of simultaneously used vasoactive drugs and tMCS devices).
Abbreviations:
Bl: baseline (admission to CSWG sites)
q6, q12, q18, q24, q48, q72: 6-hour time intervals from CS diagnosis



Journal Pre-proof

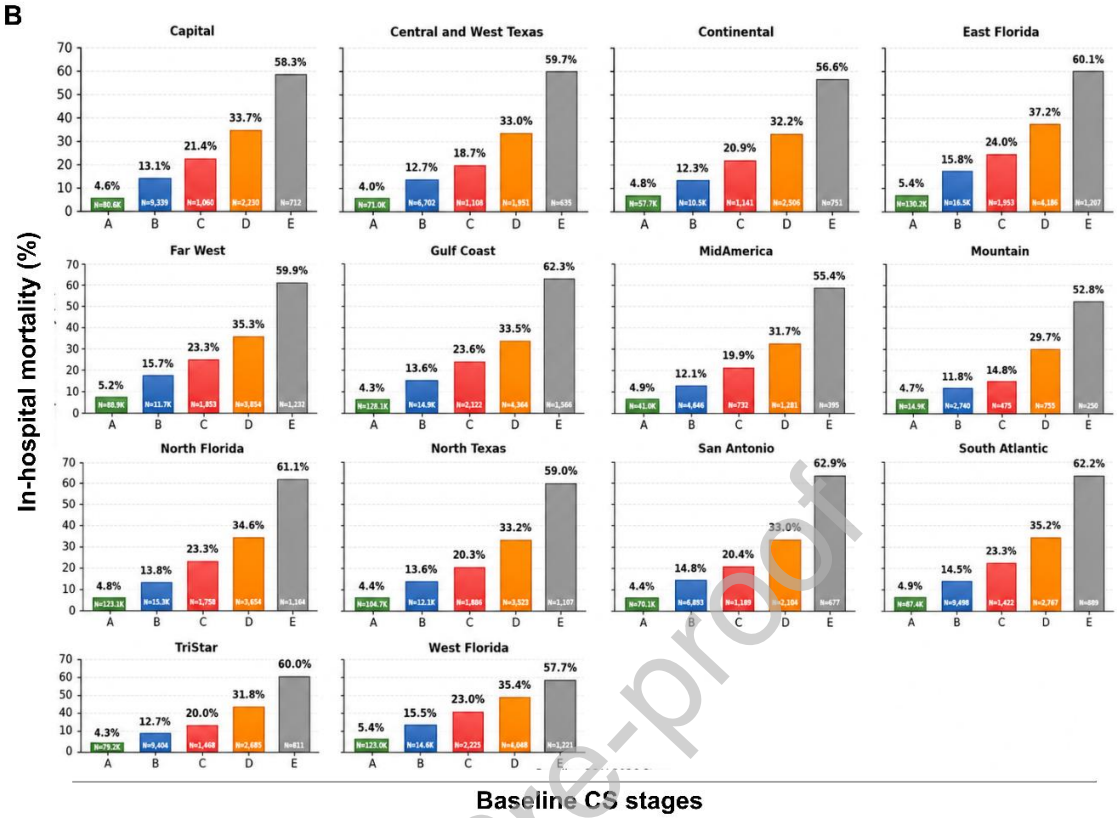
Figure 3: Application of the updated CSWG criteria. The updated CSWG criteria were

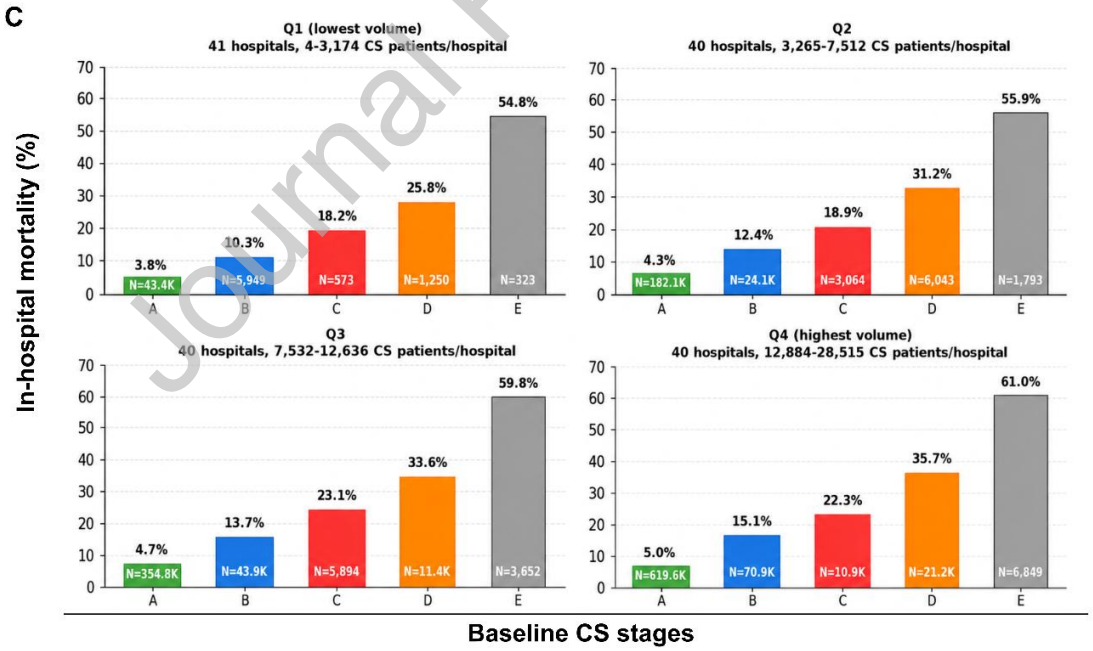
applied to the entire CSWG cohort. **A.** Proportions in the first 72 hours post CS diagnosis. **B.** In-hospital mortality at each assessed time point in all patients by the updated 2026 system (left) and the 2022 system (right). Numbers in graph: number of patients in each subgroup. Time 0: admission to CSWG sites (baseline).



4a

Journal Pre-proof

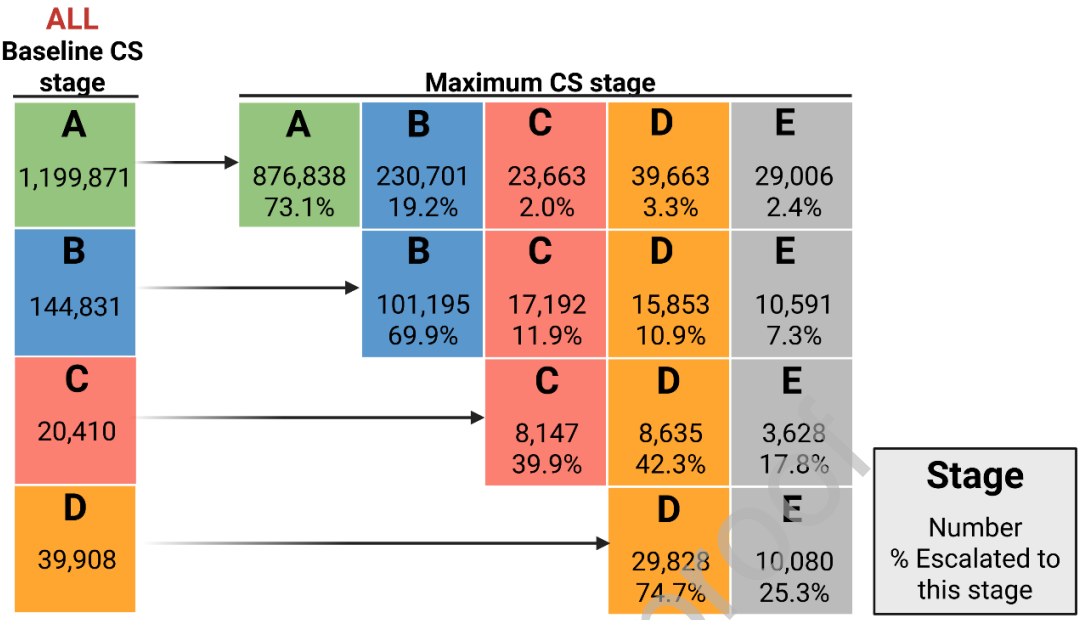




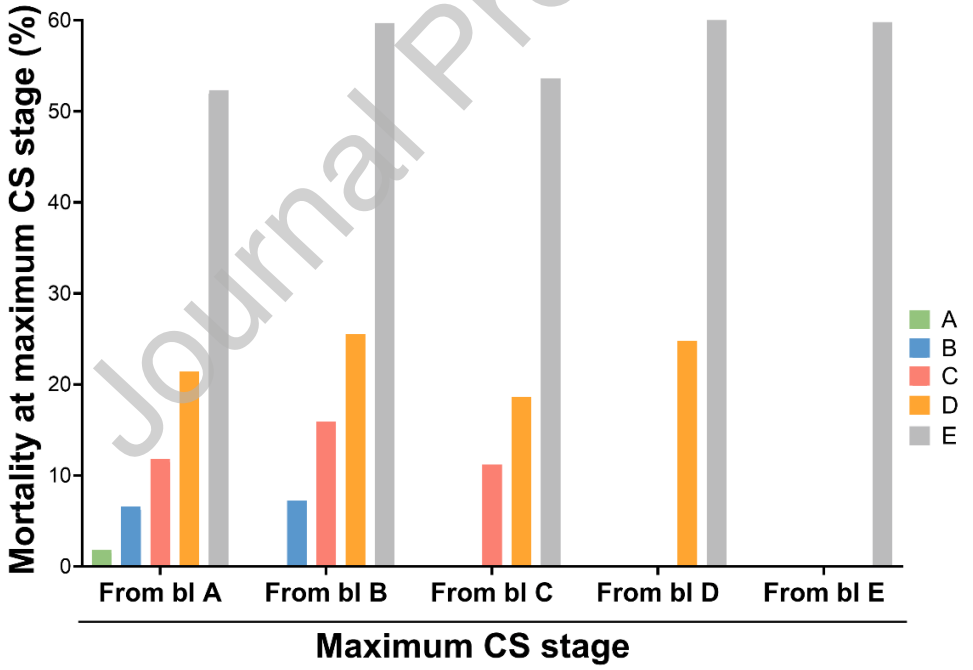
4b-c

Journal Pre-proof

D

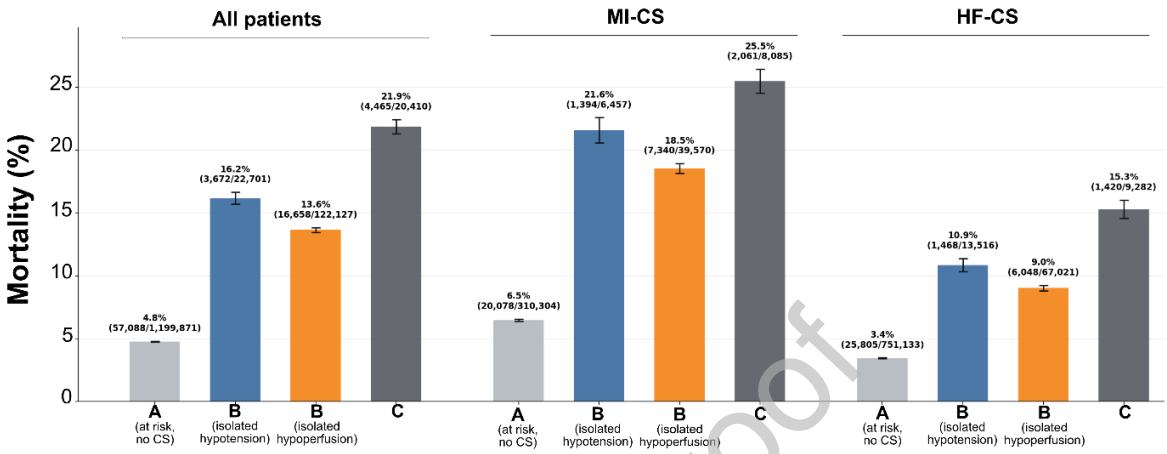


E



4d-e

Journal Pre-proof



4f

Figure 4: External application. The updated CSWG stages were applied in 1,417,637 patients admitted at 161 hospitals in 20 U.S. states (HCA Healthcare). **A.** In-hospital mortality of all patients, classified by the 2026 stages (left panel) or 2022 stages (right panel). **B.** In-hospital mortality of patients stratified by geographical locations. **C.** In-hospital mortality of patients stratified by hospital volumes. **D.** Proportions of patients progressing from baseline to maximum CS stages. **E.** Mortality of patients progressing from baseline stages A-E to maximum stages throughout hospitalization. **F.** Mortality of patients with isolated hypotension (SBP <90mmHg) vs. isolated, mild lactate elevation (>2-5mmol/L) vs. both hypoperfusion and hypotension. Bl = baseline. HF-CS: heart failure-related cardiogenic shock. MI-CS: myocardial infarction-related cardiogenic shock.

Journal Pre-proof

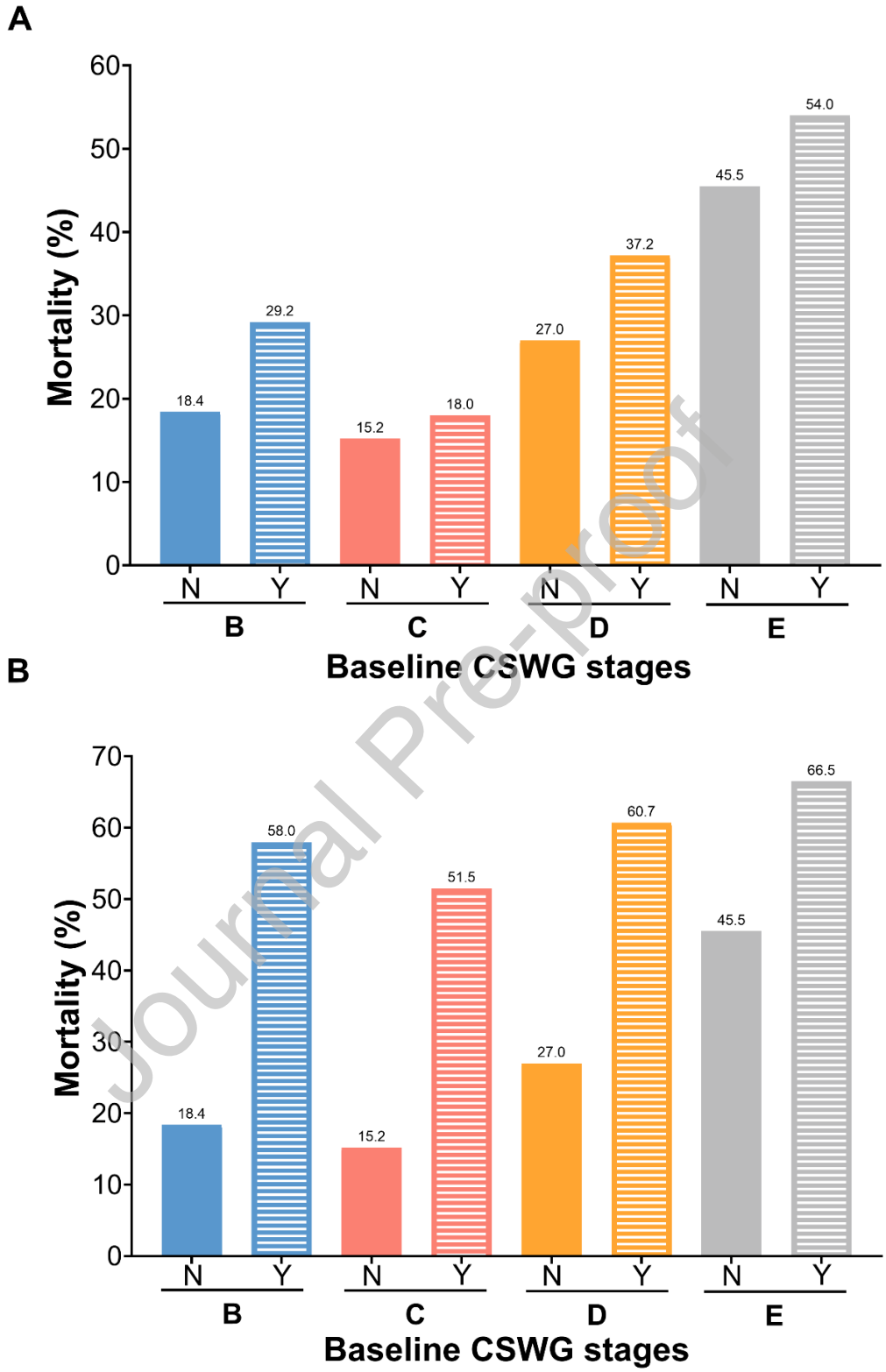
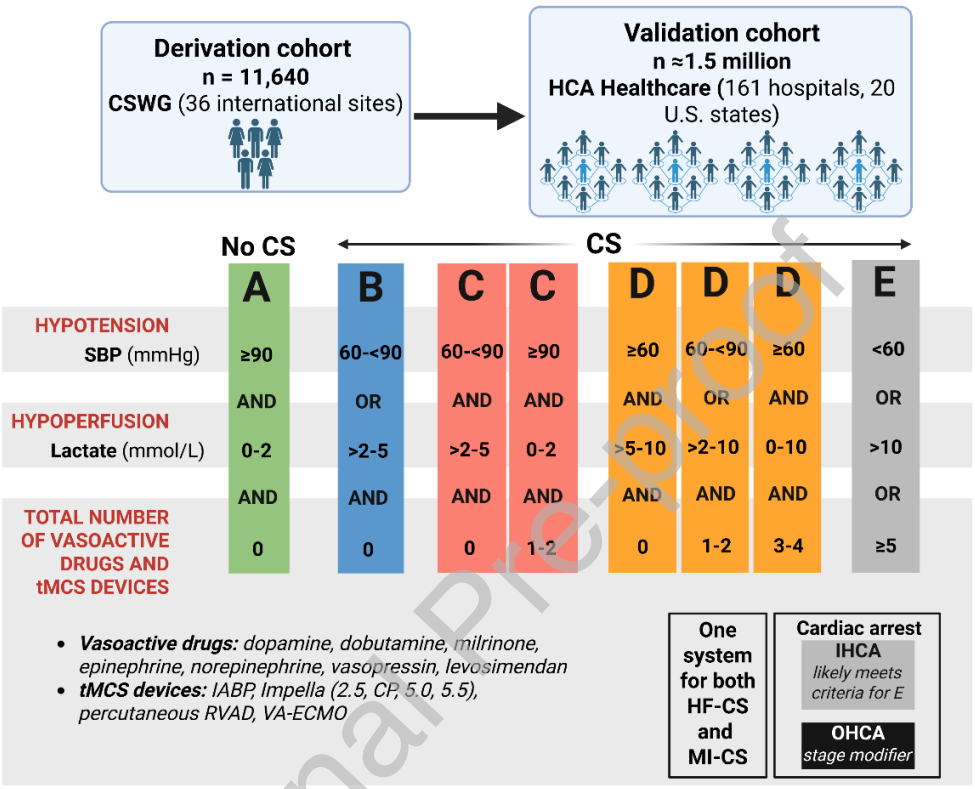
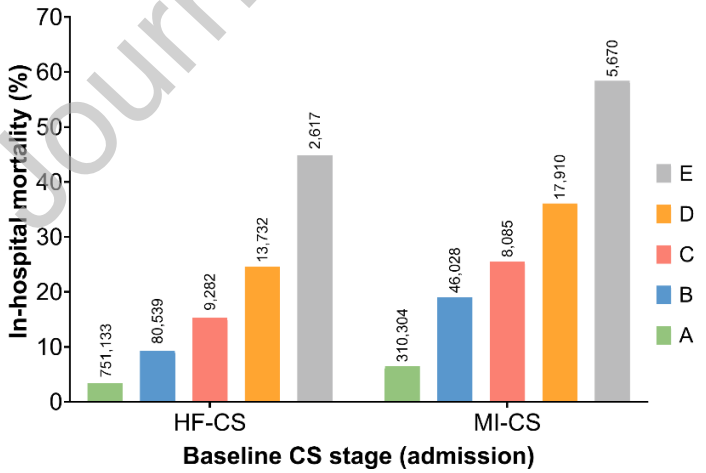


Figure 5: Patients with cardiac arrest in CSWG cohort. The updated CSWG stages were

applied to patients with and without cardiac arrest. **A.** Mortality with and without out-of-hospital cardiac arrest (OHCA). **B.** Mortality with and without in-hospital cardiac arrest (IHCA).
Y = arrest present; N = arrest absent.

CENTRAL ILLUSTRATION: Data-Driven Evolution of the CSWG Staging System





Central Illustration: The updated CSWG staging system was derived from CSWG cohort and externally applied in HCA Healthcare dataset. Mortality incrementally increased from stages A

to E in HF-CS and MI-CS cohort.

Abbreviations: CSWG: the Cardiogenic Shock Working Group, ECMO: extracorporeal membrane oxygenation, HF-CS: heart failure-related cardiogenic shock, IABP: intra-aortic balloon pump, IHCA: in-hospital cardiac arrest, MI-CS: myocardial infarction-related cardiogenic shock, OHCA: out-of-hospital cardiac arrest, RVAD: right ventricular assist device, tMCS: temporary mechanical circulatory support.

TABLES

Table 1: Baseline characteristics. Demographics, medical history, laboratory and hemodynamic values on admission, and in-hospital mortality of CSWG cohort stratified by HF-CS or MI-CS.

	All (N=11640)		**HF-CS (N=7039)**		**MI-CS (N=3040)**		***p-values***
	N	**N (%) or Median [Q1-Q3]**	**N**	**N (%) or Median [Q1-Q3]**	**N**	**N (%) or Median [Q1-Q3]**	
Demographics							
Male	8040 (69.1)		4930 (70)		2194 (72.2)		0.06
Age (years)	11224	63 [53-71]	6837	62 [51-70]	2982	66 [58-74]	**<0.001**
Race							**<0.001**
White	7476 (64.2)		4441 (63.1)		2075 (68.3)		
Black	2135 (18.3)		1592 (22.6)		233 (7.7)		
Asian	513 (4.4)		240 (3.4)		221 (7.3)		
Other	740 (6.4)		330 (4.7)		259 (8.5)		
Body mass index (kg/m^2)	11384	27.9 [24.1-32.7]	6944	27.8 [23.9-32.8]	2993	27.7 [24.3-31.9]	0.29
Medical history							
Hypertension	7463 (64.1)		4504 (64)		1990 (65.5)		**0.04**
Diabetes	4650 (39.9)		2794 (39.7)		1334 (43.9)		**<0.001**
Atrial fibrillation/flutter	3671 (31.5)		2863 (40.7)		389 (12.8)		**<0.001**
Chronic kidney disease	3490 (30)		2556 (36.3)		541 (17.8)		**<0.001**
Peripheral vascular disease	1114 (9.6)		666 (9.5)		314 (10.3)		0.13
COPD	1554 (13.4)		1047 (14.9)		309 (10.2)		**<0.001**
CVA/TIA	1464 (12.6)		969 (13.8)		323 (10.6)		**<0.001**
Valvular disease	2382 (20.5)		1776 (25.2)		249 (8.2)		**<0.001**
ICD	2570 (22.1)		2219 (31.5)		145 (4.8)		**<0.001**
Transfer	5715 (49.1)		3192 (45.3)		1878 (61.8)		**<0.001**
Prior PCI	2268 (19.5)		1410 (20)		647 (21.3)		0.10
Prior CABG	1153 (9.9)		795 (11.3)		228 (7.5)		**<0.001**
Laboratory values							
ALT (/L)	9749	43 [22-135]	5976	38 [20-120]	2603	61 [28-166]	**<0.001**
BUN (mg/dL)	11111	28 [19-44]	6740	31 [21-48]	2971	24 [17-37]	**<0.001**
Lactate (mmol/L)	8145	2.3 [1.4-4.4]	4765	2.1 [1.4-3.8]	2421	2.6 [1.5-5.3]	**<0.001**
HCO3 (mEq/L)	10342	22 [19-25]	6296	23 [20-26]	2805	21 [18-24]	**<0.001**
Creatinine (mg/dL)	11175	1.5 [1.1-2.2]	6767	1.5 [1.1-2.2]	2998	1.3 [1-2]	**<0.001**
pH	5953	7.4 [7.3-7.4]	3087	7.4 [7.3-7.4]	2000	7.3 [7.2-7.4]	**<0.001**
International normalized ratio	7941	1.4 [1.2-1.8]	4527	1.5 [1.2-2.1]	2392	1.2 [1.1-1.5]	**<0.001**
Total bilirubin	9923	1 [0.6-1.7]	6042	1.2 [0.7-2]	2700	0.7 [0.5-1.2]	**<0.001**
Hemodynamics							

Journal Pre-proof

Heart rate (bpm)	11156	93.2±23.9	6758	94.4±23.8	2969	91.3±23.4	**<0.001**
Systolic blood pressure (mmHg)	11027	110 [97-126]	6692	108 [96-123]	2931	112 [97-131]	**<0.001**
Mean arterial pressure (mmHg)	8722	83 [73-94]	5323	83 [74-94]	2341	83 [73-96]	0.17
Right atrial pressure (mmHg)	1362	13 [9-18]	714	14 [9-18]	557	12 [9-16]	**0.009**
Systolic pulmonary artery pressure (mmHg)	1475	42 [32-53]	754	46 [35-57]	619	38 [30-48]	**<0.001**
Diastolic pulmonary artery pressure (mmHg)	1459	23 [17-29]	748	25 [18-31]	608	21 [16-27]	**<0.001**
Mean pulmonary artery pressure (mmHg)	1455	29.3 [22.7-37]	747	32.3 [24.3-39.7]	606	27.3 [21.3-33.7]	**<0.001**
Pulmonary artery wedge pressure (mmHg)	462	22 [16-28]	249	22 [17-29]	187	21 [15-28]	0.08
Cardiac output (L/min)	872	4 [3.2-5.3]	460	3.9 [3.1-5]	360	4.3 [3.2-5.6]	**0.01**
Cardiac index (L/min/m^2)	867	2.1 [1.6-2.6]	459	2 [1.6-2.5]	357	2.2 [1.6-2.9]	**0.005**
Outcomes							
In-hospital mortality	3350 (28.8)		1702 (24.2)		1127 (37.1)		**<0.001**

Abbreviations: ALT: alanine transaminase, BUN: blood urea nitrogen, CABG: coronary artery bypass grafting, COPD: chronic obstructive pulmonary disease, CVA: cerebrovascular accident, HF-CS: heart failure-related cardiogenic shock, ICD: internal cardioverter-defibrillator, MI-CS: myocardial infarction-related cardiogenic shock, OHCA: out-of hospital cardiac arrest, PCI: percutaneous coronary intervention, TIA: transient ischemic attack,

Table 2: Updates to the CSWG staging system. Prior criteria for CS staging are compared to the current, updated criteria.

Criterion	Variable	Threshold	
		2022 CSWG	**2026 CSWG**
Hypotension	Systolic blood pressure (mmHg)	60 - 90 <60	≥90 60 - <90 <60
	Mean arterial pressure (mmHg)	50 - 65 <50	Not needed
Hypoperfusion	Lactate (mmol/L)	2 - 5 >5 - 10 >10	0 – 2 >2 – 5 >5 – 10 >2 -10 0 – 10 >10
	pH	<7.2	Not needed
	ALT (U/L)	200 – 500 >500	Not needed
Treatment intensity	Total number of vasoactive drugs and tMCS devices	Stage C: 1 Stage D: 2-5 Stage E: ≥3	Stage C: 1-2 Stage D: 3-4 Stage E: ≥5
Cardiac arrest	OHCA	Automatic stage E	Stage modifier
	IHCA	Not addressed	Automatic stage E

Journal Pre-proof

Abbreviations: ALT: alanine transaminase, IHCA: in-hospital cardiac arrest, OHCA: out-of hospital cardiac arrest, tMCS: temporary mechanical circulatory support

Table 3: Reclassification between 2022 and 2026 CS staging systems. Number of patients reclassified, and associated in-hospital mortality. Those missed by the 2022 system (no stage assignment) are now classified in the 2026 system.

2022 CS stages	2026 CS stages	N	Proportion (%)	Death (n)	Mortality (%)	Total reclassified (%)
A	A	953	100.0	193	20.3	0
B	A	113	9.9	22	19.5	9.9
	B	1027	90.1	251	24.4	
C	B	16	2.0	4	25	2.0
	C	781	98.0	133	17	
D	A	24	0.8	3	12.5	28.4
	B	109	3.7	25	22.9	
	C	597	20.2	141	23.6	
	D	2117	71.6	686	32.4	
	E	109	3.7	50	45.9	
E	A	72	3.8	18	25	56.5
	B	107	5.6	34	31.8	
	C	155	8.1	42	27.1	
	D	750	39.1	337	44.9	
	E	836	43.5	472	56.5	
No stage assignment	A	136	71.6	35	25.7	100
	C	54	28.4	9	16.7	

Journal Pre-proof



Author headshot photograph

Journal Pre-proof

Conflict of Interest and Financial Disclosure: This work was supported by institutional grants from Abiomed Inc, Boston Scientific Inc, Abbott Laboratories, Getinge Inc, and LivaNova Inc to Tufts Medical Center. The sponsors had no input on collection, analysis, and interpretation of the data, nor in the preparation, review, or approval of the manuscript. Dr. Kapur has received consulting honoraria and institutional grant support from Abbott Laboratories, Abiomed Inc, Boston Scientific, Medtronic, LivaNova, Getinge, and Zoll, and is an employee of Johnson & Johnson. Dr. Ton reports consulting fee from Abiomed Inc., Johnson & Johnson. Dr. Kanwar has served on the advisory board for Abiomed Inc., Abbott Laboratories and CorWave. Dr. Sinha serves as the NPI for the OASIS AMICS trial (J&J MedTech) and PI for the HALO Shock trial (Abbott). Dr. Garan has served as a consultant for NuPulseCV, has been on the scientific advisory board for Abiomed, and is a recipient of research support from Verantos and Abbott. Dr. Hernandez-Montfort has served as a consultant for Abiomed Inc. Dr. Abraham has served as a

consultant for Abbott Laboratories and Abiomed Inc, and is an employee of Edwards Lifesciences. Dr. Vorovich has been a consultant and served on the speaker bureau, advisory board, and steering committee for Abiomed, Inc., a speaker for Abbott Laboratories, and served on the advisory board for Novo Nordisk A/S. Dr. Nathan has received consulting honoraria from Abiomed, Getinge, and CSI. Dr. Hall has served as a consultant to Abiomed, Abbott, and Medtronic. Dr. Mahr has served as a consultant to Abbott, Abiomed, and Syncardia. Dr. Burkhoff has received an unrestricted, educational grant from Abiomed Inc. Dr. Yeh has received research grants and consulting fees from Abbott Vascular, Boston Scientific, and Medtronic, and consulting feels from Magenta Medical and Supira Medical. Dr. Møller has received an unrestricted, research grant from Johnson and Johnson Med Tech Heart Recovery and Novo Nordic Foundation and has served as advisor for Boston Scientific and Magenta. All other authors have reported that they have no relationships relevant to the contents of this paper to disclose.

Journal Pre-proof